

Cross-Age Face Verification from Mined Same-Person Social-Media Pairs: A Naturally-Supervised Longitudinal Signal

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Abstract—Cross-age face verification lacks longitudinal training data. We mine weak same-person pairs from social-media “then/now” posts using caption parsing and embedding-based person clustering, without manual identity or age labels. On an endpoint-age-matched FG-NET protocol, fine-tuning FaceNet raises large-gap (≥ 25 -year) ROC-AUC from 0.802 to 0.8490 ± 0.0016 across three seeds; the mean gain is $+0.0467 \pm 0.0016$. Each checkpoint’s paired subject-bootstrap 95% gain interval is positive on the same subjects; these are not intervals for the training-seed population. The larger $+0.112$ gain from the original random-impostor protocol is historical. Source-bound LFW AUC improves, but accuracy and low-FAR gain intervals include zero. CALFW/CACD-VS AUC changes little, AgeDB-30 decreases by 0.008, and an internal endpoint-matched test gains AUC but loses low-FAR TAR. Tested objectives have similar historical point estimates, not demonstrated equivalence. A fixed-epoch within-VK control favors long-gap pairs but has unequal total image budgets; matched-budget retraining remains incomplete. Thus causal source superiority, benchmark independence and a strong-backbone mechanism are not established. Bidirectional VK–Reddit transfer and the tested FRAN controls provide bounded supporting evidence, not population-wide transfer or superiority over synthesis methods in general.

Index Terms—Cross-age face recognition, age-invariant representation, naturally/weakly-supervised identity, dataset curation, shortcut learning, fairness, calibration.

I. INTRODUCTION

CROSS-AGE face verification arises in archival retrieval, account recovery, and authorized search for missing persons. The challenge lies not in discriminating among arbitrary individuals but in distinguishing *the same person across age* from *different people of similar age and appearance*. Annotating longitudinal identity pairs (one individual across many years) is costly, and such data remain scarce, which constrains supervised approaches; recent work continues to identify both the shortage of cross-age supervision and cross-dataset generalization as central problems [?], [8].

Hypothesis. A multi-photo “then/now” post showing one person at different ages yields $\binom{n}{2}$ positive identity pairs without manual annotation; a caption stating ages adds a secondary age anchor. The same-person pairing is the operative signal (an ablation later isolates the age anchor at a marginal $+0.009$); this

is a scalable, naturally-supervised signal of cross-age identity, and fine-tuning a weak recognizer on it should transfer to external benchmarks.

Contributions (bounded). (1) A weakly/naturally-supervised (no manual id/age labels) longitudinal signal mined from social posts—same-person identity pairs, with caption age anchors only a secondary cue. (2) A versioned curation pipeline with automatic component checks, quality/sensitivity analysis, and the finding that naive positive counts are inflated about $2\times$ by near-duplicate frames; human validation and approved data access remain pending. (3) A controlled *evaluation protocol* (one weak trainable backbone, frozen vs. fine-tuned, external evaluation) measuring the utility of training on mined pairs while holding network capacity fixed; causal attribution to longitudinal pairing requires additional matched-data controls. (4) A bounded account of where the gain does and does not persist: comparisons across tested objectives, backbones, source communities and external benchmarks, plus explicit low-FAR trade-offs. The effect of longitudinal pairing relative to matched-source controls remains to be established.

We emphasize that the novelty is data-centric rather than architectural—a new *origin of supervision*—and should be evaluated accordingly.

II. RELATED WORK AND POSITIONING

Age disentanglement and synthesis. OE-CNN (orthogonal embedding decomposition) [1], DAL (decorrelated adversarial learning) [2], MTLFace (recognition, age synthesis and attention decomposition) [3], and CDD-AFL (class-wise domain decoupling with adversarial feature learning) [?] learn age-invariant features from manually labeled identity/age data. Face-aging augmentation is a second route, but controlled studies find only partial recognition gains [10].

Contrastive and data-centric supervision. CACon adds a synthesized cross-age view to a multi-positive contrastive objective [8]; OrdCon explicitly models ordinal age progression for cross-dataset generalization [9]. HMLF combines margin and contrastive losses without age labels for $1:N$ retrieval [?]; its mAP/Rank- k protocol is complementary, not a 1:1 verification baseline. Generic SimCLR-style augmentation [11] lacks identity anchors across time. We instead mine the structure “one post = one person at different ages” as a naturally-supervised positive-pair generator.

Scope and evaluation. Direct cross-age methods above are separated from LARNet/LARNeXt, which model a *pose*-induced feature residual [?], [?] and motivate our diagnostics

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Non-sensitive code, configurations and aggregate statistics are available at <https://github.com/deneal123/ChildVSAdult>; no faces, biometric templates or row-level private-corpus protocol are released.

only by analogy. The standard evaluation datasets—CACD [4], MORPH [5], AgeDB [6], FG-NET [7], and CALFW [12]—mix verification, identification, and retrieval protocols, so published headline numbers cannot establish controlled superiority. Table I positions the supervision mechanisms; the supplement separates published-number comparisons from our common-protocol reproductions.

We report the canonical verification suite (AgeDB-30, CALFW, CACD-VS and FG-NET; CALFW is an age-gapped LFW protocol [13]). The goal is *not* absolute leaderboard superiority: we use a weak backbone for clean attribution and quantify the transfer from a naturally-supervised source; Sec. V-F then compares complete methods under one controlled protocol.

III. DATA: NATURALLY-ANCHORED LONGITUDINAL PAIRS

Source. Two public VK then/now communities (Russian-language context) supply candidate longitudinal views. InsightFace `buffalo_1` RetinaFace [14] and 5-point ArcFace alignment produce usability records with quality/target-selection decisions. Figure ?? summarizes recorded yields; face records include rejected-photo placeholders, not distinct processed photos.

A. Age Extraction

Captions are parsed by regular expressions (Russian patterns plus the slash format “6/18/21”, one age per photo) and GigaChat-2-Max, a Russian-oriented LLM family [?]. The frozen run uses temperature 10^{-5} , top- $p = 0$, a hashed prompt and cached responses. The model was selected for Russian-language support, not brand recognition; a final quality justification is conditional on the prepared human-gold comparison against regex and a reproducible local model. Recomputing the comparison directly from all 33,697 cached multi-photo captions (Table III) gives 78.1% exact agreement (including both-empty outputs); 15.5% return different non-empty age sets, while LLM-only and regex-only outputs account for 3.4% and 3.0%. These figures are an automatic cross-check, not accuracy against human ground truth. We do not treat either system as ground truth: a prespecified multi-annotator audit is required before resubmission, and downstream explicit age anchors contribute only marginally (Sec. V-A).

B. Persons and Leakage-Safe Split

The hierarchy is post $\rightarrow$ candidate group $\rightarrow$ recorded person $\rightarrow$ face $\rightarrow$ pair. ArcFace R50 w600k_r50 embeds crops; union-find merges groups linked by cosine ≥ 0.85 . The map covers 27,487 groups and 21,847 components; one unmapped two-face group yields 21,848 pre-curation groups from 27,488 candidates. Splits use recorded persons, not verified identities; missed merges remain possible. Independent FaceNet/CASIA yields 21,484 components (1.7% difference); blinded validation remains pending.

C. Auditing the Automatic Supervision

Automatic supervision can hide label noise. Checks cover age agreement (Table III), merging, integrity and deduplication. InsightFace `buffalo_1/genderage.onnx` supplies apparent binary gender; consistency is the majority fraction among available predictions. Of 2,919 pre-prune multi-face groups below 0.6, 2,866 remain: consistency is diagnostic, not a pruning rule or identity evidence. A prespecified held-out two-annotator audit with agreement and intervals remains pending; no human-validated precision is claimed.

D. Curation Pipeline and Its Largest Finding

Three steps: (i) apparent gender/age metadata (76.7% apparent-female); (ii) LLM integrity labels from each group’s representative post, not all constituent captions, removing 421 groups/830 faces; (iii) within-group deduplication (recipe cosine ≥ 0.97), removing 8,190 more faces. The output has 21,427 groups/40,586 faces. Clustering merges groups; it does not separate multiple people within a post. Reconstruction from the current redundant-face file is incomplete; lineage and rerun limitations are in the supplement.

Key finding. Because the number of positives is quadratic in the number of faces per group, removing near-duplicates reduced the positive count from 65,897 to 31,865 (−52%): **approximately half of the raw “positives” were duplicate frames** (a single shoot, burst, or repost) that had inflated the internal metrics. After curation, the frozen baseline on the internal test decreases (our.overall 0.856 $\rightarrow$ 0.836; our.25+ 0.705 $\rightarrow$ 0.640), exposing a harder and more representative benchmark. The external benchmarks are unaffected. This constitutes a transferable methodological observation for mining longitudinal pairs from social media.

Threshold robustness. The −52% reduction is not an artifact of the 0.97 cutoff. Re-running the near-duplicate dedup on the pre-curation groups at cosine thresholds 0.93–0.99 (Table VI) leaves 31.3k–32.4k positives across 0.93–0.97 (a stable $\approx 52\%$ reduction) and degrades gracefully only at the very strict 0.99 (−41%): the near-duplicates are dominated by near-identical frames (cos ≈ 1), so the finding does not hinge on the exact cutoff.

E. Dataset Datasheet (Summary)

Provenance, rights and governance fields are stored per item (source, origin URL, collection date, license/consent status, allowed use, retention policy, deletion status, review status). Collection: public VK walls. Intended use: research on consent-based cross-age matching. Out of scope: mass identification, surveillance, de-anonymization. Minors: the child $\leftrightarrow$ adult regime is in scope only under an explicit legal/ethical framework; per-item review status is tracked. A datasheet [15], controlled-access procedure and deletion workflow are drafted, but no biometric artifact will be distributed until institutional and legal review approves their final form.

IV. METHOD, PROTOCOL AND STATISTICS

Human subjects and consent. The data are photographs of identifiable people. Individual informed consent for participation and publication was *not* obtained; Sec. VIII explains

TABLE I
POSITIONING VS. PRIOR WORK.

Work	Supervision	Mechanism	Our difference
OE-CNN (2018) [1]	labeled id+age	orthogonal feature decomposition	no manual id/age labels, not decomposition
DAL (2019) [2]	labeled id+age	adversarial id/age factorization	mined pairs + simpler objective
MTLFace (2021) [3]	labeled id+age	recognition + age synthesis (multi-task)	weaker pipeline; novel supervision origin
CACCon (2024) [8]	semi-sup. + synthetic	contrastive on synthesized cross-age	natural cross-time views; controlled reproduction
Synthetic Ageing (2024) [10]	synthetic	train on aged images	we make <i>real</i> mining the thesis
OrdCon (2025) [9]	age-ordered contrastive	order-enhanced features	mine natural pairs, no explicit age order

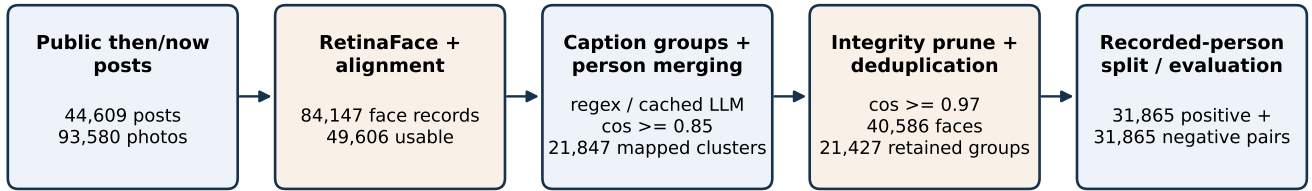


Fig. 1. Recipe and checksum-bound recorded yields. Merging precedes integrity pruning/deduplication. Thresholds describe the recipe, not historical preprocessing replay. The prepared five-target audit (400 items each) still requires two independent annotators and adjudication.

TABLE II
RECORDED SCALE BEFORE/AFTER CURATION. RAW GROUPS ARE MAPPED CLUSTERS; ONE FALLBACK GROUP ALSO ENTERS CURATION (SEC. III-B).

Stage	Posts	Photos	Face records	Usable faces	Groups	Positives
Raw mined (2 communities)	44,609	93,580	84,147	49,606	21,847	65,897
After curation	—	—	—	40,586	21,427	31,865

TABLE III
AGE-EXTRACTION AUDIT: LLM VS. REGEX (33,697 CAPTIONED MULTI-PHOTO POSTS).

Category	<i>n</i>	Share
Agree (same ages or both empty)	26,312	78.1%
LLM-only ages	1,162	3.4%
Different non-empty ages	5,207	15.5%
Regex-only ages	1,016	3.0%

TABLE IV
LLM GROUP-INTegrity CLASSIFICATION (33,697 POSTS).

Category	<i>n</i>	Share
Single (one person across ages)	31,275	92.8%
Multi-person (different people)	2,034	6.0%
Unknown	336	1.0%
Meme / collage	52	0.2%

the current basis and safeguards. A formal institutional determination has not yet been obtained and is a prerequisite for resubmission; the final manuscript must report the committee’s actual decision rather than a self-assessed exemption. **Attribution protocol.** Comparing an adapter on top of a *frozen* strong

TABLE V
DEDUP-THRESHOLD SENSITIVITY: NEAR-DUPLICATE REMOVAL ALONE, ON THE PRE-CURATION GROUPS (THE SEPARATE GROUP-INTegrity PRUNE REMOVES A FURTHER ≈ 500 POSITIVES TO REACH THE HEADLINE 31,865). STABLE ACROSS 0.93–0.97.

Dedup cos	Near-dup faces	Positives	Reduction
0.93	8,653	31,304	−52.5%
0.95	8,564	31,611	−52.0%
0.97 (used)	8,304	32,360	−50.9%
0.99	6,274	38,565	−41.5%

ArcFace against frozen ArcFace is invalid (the adapter merely deforms near-perfect embeddings). Instead we use *one weak trainable backbone* (FaceNet InceptionResnetV1 [16], CASIA-WebFace) and compare (a) frozen $\rightarrow$ benchmark metric vs. (b) soft fine-tuning of the head on our pairs $\rightarrow$ benchmark metric. Training uses a margin contrastive loss on aligned crops. *Evaluation is on external benchmarks* (LFW, AgeDB-30, CALFW, CACD-VS and FG-NET) to assess transfer; the internal test (our. *) is a secondary in-domain diagnostic. Backbone strength is varied (Sec. V-C) to map where fine-tuning helps. This isolates a fine-tuning comparison, not the causal effect of the longitudinal source without matched-source controls.

Backbones. FaceNet (weak); ArcFace iResNet-50 [17] (CASIA-FaceV5; weak, different architecture); AdaFace IR-50/IR-101 [18] (WebFace; strong; a clean depth control); ArcFace iResNet-100 (strong).

Benchmarks and metrics. LFW (source-bound FaceNet official View-2 accuracy and ROC-AUC, separate from historical interleaved-cache results), AgeDB-30 and CALFW (aligned pairs; ROC-AUC), FG-NET (ROC-AUC overall and on the large-gap ≥ 25 -year subset; historical random impostors are separate from endpoint-age-matched negatives), and the official 10-fold CACD-VS pair protocol (4,000 pairs). CACD-VS reports fold accuracy, ROC-AUC, EER and TAR at FAR 1% and 0.1% with bootstrap CIs. AgeDB-30, CALFW and CACD-VS provide larger standard cross-age checks. Frozen and fine-tuned models share aligned arrays; supplied AgeDB/CALFW crops are used as provided. The new LFW table and conditional subject intervals are in the supplement: AUC improves, but accuracy and low-FAR gain intervals include zero. We distinguish *threshold-free* metrics (ROC-AUC) from *threshold-based* ones (accuracy, TAR@FAR).

Statistics. Our corrected *primary endpoint* is endpoint-age-matched FG-NET large-gap (≥ 25 -year) ROC-AUC, with the null hypothesis that fine-tuning on the mined pairs does not improve it over the frozen baseline (gain ≤ 0). The original random-impostor protocol is retained as a secondary historical comparison, not as the headline test; AgeDB-30, CALFW and the internal cross-age test are secondary endpoints, and easy LFW accuracy is monitored as a forgetting control. The headline gain is mean $\pm$ std over three seeds (seed variance captures training stochasticity, not full evaluation uncertainty); paired subject-bootstrap intervals account for FG-NET identity reuse. The fairness audit uses a paired percentile bootstrap (1000 resamples, resampling pairs and recomputing both models on the same resample) for the *gain*, accounting for frozen/tuned correlation. Table VIII reports bootstrap CIs, EER and TAR@FAR for the original protocol comparisons (the internal test additionally uses an identity-level, person-resampling bootstrap), and a Holm correction is applied across the stratified fairness tests.

Curated dataset. All headline experiments are on the curated data of Sec. III-D: 21,427 person groups, 40,586 faces, 31,865 positives; split train 22,179 / val 4,579 / test 5,107 with balanced negatives per split.

V. RESULTS

A. Main Effect and Supervision-Source Ablation

The weak FaceNet, softly fine-tuned on our pairs, raises FG-NET large-gap ROC-AUC **0.802** $\rightarrow$ **0.849** under endpoint-age-matched impostors (Table ??). Under the original random-impostor protocol it rises 0.736 $\rightarrow$ 0.848 (+0.112 $\pm$ 0.001), while easy LFW accuracy increases (+0.004), AgeDB-30 declines and CALFW changes little (historical comparisons in the supplement). A pre-clean ablation tests the contribution of explicit caption-derived age anchors among same-post pairs: these add only +**0.009** on the original 25+ bucket. This does not by itself isolate the source effect. The method is *weakly supervised by naturally occurring same-post co-occurrence* (not

by manual identity/age labels), with explicit age supervision contributing only marginally in this ablation.

The historical comparisons show low training-seed variability (std ≤ 0.003), which is not a confidence interval over subjects. Under that protocol, the internal cross-age gain grows after curation (our.25+ +0.195): the frozen baseline falls to 0.640 while +pairs recovers to 0.835. CACD-VS is effectively unchanged in threshold-free AUC but exposes a trade-off at fixed operating points: 10-fold accuracy decreases from 0.9793 to 0.9700 $\pm$ 0.0027 and EER rises from 0.0220 to 0.0313 $\pm$ 0.0013.

Bootstrap confidence intervals (seed 42) describe the original random-impostor endpoint: FG-NET large-gap rises from 0.736 [0.70, 0.78] to 0.848 [0.82, 0.88] with *non-overlapping* 95% CIs; the equal-error rate falls by about 35% (0.335 $\rightarrow$ 0.219) and TAR@FAR=1% **nearly doubles** (0.21 $\rightarrow$ 0.39). On the internal 25+ test the gain holds under the more conservative *identity-level* bootstrap (resampling persons, not pairs): +pairs 0.838 [0.80, 0.88] vs. frozen 0.640 [0.59, 0.69], an interval wider than the pair-level one as expected when pairs share identities. External operating points and pair-level intervals are in Table VIII; the internal identity-level interval is given in text. Because bootstrap-CI overlap ignores the samples the two models share, we test the original random-impostor endpoint with DeLong's test for two *correlated* ROC curves [19]: on FG-NET large-gap $z = 9.3$, $p = 1.4 \times 10^{-20}$ (Δ AUC 95% CI [+0.088, +0.135]), and on the internal 25+ test $p = 1.7 \times 10^{-34}$; a 10^4 -fold paired permutation test agrees ($p < 10^{-4}$). The same test confirms the gain is *targeted*: AgeDB-30 shows a small but significant decrease (−0.008, $p = 4 \times 10^{-6}$) and CALFW is statistically unchanged ($p = 0.95$).

Endpoint-age-matched protocol check (three seeds). Keeping one endpoint-age-matched negative per retained positive changes the FG-NET 25+ comparison to 215 positive and 215 negative pairs, with identical age-gap distributions (age-gap-only ROC-AUC 0.500). FaceNet ROC-AUC is 0.802 frozen and 0.8490 $\pm$ 0.0016 after fine-tuning; the mean gain is +0.0467 $\pm$ 0.0016. Each seed's paired subject-bootstrap 95% CI excludes zero (seed 42: [+0.006, +0.091]); seed-42 leave-one-subject-out gains are [+0.035, +0.052]. The three intervals reuse the same FG-NET subjects and are not independent replications. This check does not inherit the old protocol's +0.112 effect size or DeLong p -value; comparator re-evaluation and full training provenance remain pending.

B. Real Pairs vs. Tested Re-Aging Controls

Replacing real positives with synthetically aged versions—a domain proxy and a real re-aging network (FRAN [20])—underperforms real pairs on the curated data, and at FRAN's default *wide-gap* setting even *degrades* below frozen: FG-NET large-gap 0.727 (**below frozen 0.736**) and our.25+ 0.549 (below frozen 0.640), whereas real pairs yield +0.112 / +0.198 (Table IX). These controls generate age changes from an input image rather than observing longitudinal variation; their scores do not identify a unique failure mechanism. A native-resolution sensitivity check yields similar point estimates: re-cropping a

TABLE VI

CORRECTED FG-NET ENDPOINT-AGE-MATCHED EVALUATION (FACE NET). MEANS AND STANDARD DEVIATIONS COVER THREE TRAINING SEEDS; THE PAIRED SUBJECT-BOOTSTRAP INTERVAL IS FOR SEED 42, NOT FOR THE SEED MEAN. ALL SEEDS REUSE THE SAME SUBJECTS. LEGACY CHECKPOINT TRAINING MANIFESTS ARE ABSENT; TRAIN-BENCHMARK IDENTITY OVERLAP REMAINS UNDER REVIEW.

Stratum	Pairs / subjects	Frozen AUC	Tuned AUC	Gain	Gain 95% CI (s42)
25+ years (primary)	430 / 75	0.8023	0.8490 $\pm$ 0.0016	+0.0467 $\pm$ 0.0016	[+0.0060, +0.0906]
Overall	5308 / 82	0.8870	0.9202 $\pm$ 0.0005	+0.0332 $\pm$ 0.0005	[+0.0149, +0.0504]

TABLE VII

ORIGINAL RANDOM-IMPOSTOR PROTOCOL: OPERATING POINTS AND PAIR-LEVEL 95% BOOTSTRAP CIs (FACE NET FROZEN VS. +PAIRS, SEED 42). ALL VALUES ARE ROC-AUC UNLESS NOTED; LFW IS SHOWN AS ROC-AUC FOR CONSISTENCY (ITS HISTORICAL 10-FOLD ACCURACY IS IN THE SUPPLEMENT). THE INTERNAL IDENTITY-LEVEL INTERVAL IS REPORTED IN THE TEXT, NOT THIS TABLE.

Benchmark	Frozen AUC [95% CI]	+pairs AUC [95% CI]	EER (fr. $\rightarrow$ +p.)	TAR@1%	TAR@0.1%
FG-NET large-gap	0.736 [0.70, 0.78]	0.848 [0.82, 0.88]	0.335 $\rightarrow$ 0.219	0.209 $\rightarrow$ 0.386	0.074 $\rightarrow$ 0.154
FG-NET overall	0.896 [0.89, 0.90]	0.922 [0.92, 0.93]	0.186 $\rightarrow$ 0.151	0.502 $\rightarrow$ 0.562	0.315 $\rightarrow$ 0.308
AgeDB-30	0.953 [0.95, 0.96]	0.945 [0.94, 0.95]	0.115 $\rightarrow$ 0.126	0.525 $\rightarrow$ 0.510	0.285 $\rightarrow$ 0.309
CALFW	0.948 [0.94, 0.95]	0.948 [0.94, 0.95]	0.116 $\rightarrow$ 0.117	0.580 $\rightarrow$ 0.624	0.213 $\rightarrow$ 0.320
CACD-VS	0.995 [0.992, 0.997]	0.994 [0.992, 0.996]	0.022 $\rightarrow$ 0.031	0.963 $\rightarrow$ 0.946	0.893 $\rightarrow$ 0.884
LFW (AUC)	0.983 [0.98, 0.99]	0.986 [0.98, 0.99]	0.038 $\rightarrow$ 0.037	0.939 $\rightarrow$ 0.940	0.814 $\rightarrow$ 0.816

TABLE VIII

REAL VS. TESTED SYNTHETIC POSITIVES (FACE NET, CURATED DATA, SEED 42, HISTORICAL RANDOM-IMPOSTOR PROTOCOL). FRAN-GM MATCHES THE SYNTHETIC AGE-GAP TO THE REAL-PAIR DISTRIBUTION (MEDIAN 9 Y); THE CRUDER DOMAIN PROXY REACHES 0.761 / 0.594. THE 0.838 INTERNAL AUC IS A SINGLE-SEED RESULT; THE SUPPLEMENT REPORTS THE HISTORICAL THREE-SEED MEAN 0.8348.

Metric	Frozen	+real	syn-FRAN	FRAN-gm
FG-NET large-gap	0.736	0.848	0.727	0.759
our.25+	0.640	0.838	0.549	0.622

train subset at 512 px directly from the raw images (vs. our stored 112-px crops, which FRAN upscales internally) and re-aging gives similar results—FG-NET large-gap 0.771 vs. 0.778 and our.25+ 0.671 vs. 0.670 on this 499-face subset—so the tested FRAN variants do not match real pairs at these resolutions; this is not an equivalence test or a statement about all synthesis methods. A gap-parity control sharpens this: FRAN ages to a wide 33–70-year gap whereas real pairs have a median gap of 9 years, and matching the synthetic gap to the real distribution lifts FRAN to **0.759** (our.25+ 0.622)—a *modest* positive just above frozen, in line with small synthetic-aging gains reported elsewhere [10]—yet real pairs (0.848) still lead by +0.089. The sub-frozen figure was thus partly a gap-mismatch artifact: real mined pairs beat even a *gap-matched* re-aging model.

C. Headroom Curve and the Strong-Backbone Regime

An earlier exploratory one-seed curve across FaceNet, ArcFace iResNet and AdaFace IR-50/IR-101 shows an overall, non-monotone decline in gain as frozen-model strength increases (Table XI; plotted in the supplement). This observation motivates—but does not replace—the planned fixed-budget three-seed learning-rate $\times$ trainable-scope $\times$ negative-type matrix. Until that matrix and its

gradient/CKA/drift/separability/age-probe diagnostics are complete, we do *not* label the behavior “saturation” or choose among ceiling, insufficient plasticity and catastrophic forgetting. The exploratory curve bounds the present claim to weak/medium backbones; it is not mechanistic evidence.

The random-impostor protocol understates the task.

Our test negatives are random cross-identity pairs, which modern backbones separate trivially: frozen AdaFace IR-101 reaches `our.25+ = 0.968`. This indicates limited random-impostor headroom at that endpoint, not that an adapter or another supervision regime cannot help. The reason is visible in the raw similarities. Median ArcFace cosine is 0.456 for positives below 10 years, 0.307 at 10–25 years and 0.232 at 25+, against 0.012 for random negatives—but **0.289** for the most similar same-gender, same-apparent-age impostor. *At a 25-year gap a person is less similar to themselves than to the most similar same-age stranger.* Re-mining the negatives accordingly (same apparent gender, $|\Delta\text{age}| \leq 5$ years, ranked by an independent miner outside the evaluated set) collapses every modern recognizer we test (Table X): frozen IR-101 falls from 0.961 to **0.376** [0.356, 0.394] at rank-1, *significantly below chance*. Automated checks find no same-cluster negatives (0/31 865) and no recorded identity-group IDs crossing splits; 99.1% of groups retained in positives are classified as single-person by the LLM (92.8% before curation). These are automated consistency checks, not human-verified identity purity: missed merges, labeling errors and train-benchmark overlaps remain under blinded review (Sec. III-D). A second miner from a different loss family reproduces the curve to within 0.03 and preserves the model ordering at every level, so the difficulty axis belongs to the task, not to the miner. Our mined pairs still help a weak backbone across the whole curve (+0.20 to +0.28 on the 25+ slice) and the distance to frozen SOTA narrows at the hard end (−0.145 at random vs. −0.070 at rank-1, paired bootstrap), but it does not close. The correct reading of Table XI is therefore that strong

TABLE IX

LOOK-ALIKE IMPOSTORS BREAK MODERN FACE RECOGNITION (25+ ROC-AUC; RANK = IMPOSTOR RANK UNDER AN INDEPENDENT MINER).

Top: FROZEN RECOGNIZERS ON THE *full* 25+ SET ($n_{POS} = 1208$, $n_{NEG} = 31\,590$), WITH NO FINE-TUNING ON MINED PAIRS; PRETRAINING OVERLAP IS UNRESOLVED. *Bottom:* THE EFFECT OF OUR SUPERVISION, NECESSARILY TEST-ONLY ($n_{POS} = 141$, $n_{NEG} = 4762$). THE BLOCKS USE DIFFERENT IMPOSTOR POOLS AND ARE *not* COMPARABLE ACROSS BLOCKS: RANK- k DIFFICULTY GROWS WITH POOL SIZE.

<i>Frozen recognizers, full 25+ set</i>			
Negatives	AdaFace IR-101	ArcFace r100	AdaFace IR-50
random	0.961	0.941	0.908
rank-50	0.727	0.677	0.560
rank-10	0.603	0.565	0.437
rank-1	0.376	0.362	0.258
<i>Our supervision, test split only</i>			
Negatives	FaceNet frozen	FaceNet +pairs	Δ
random	0.568	0.811	+0.243
rank-50	0.381	0.657	+0.276
rank-10	0.305	0.562	+0.257
rank-1	0.203	0.404	+0.201

backbones are not improved by *our* supervision—not that they lack cross-age headroom, and not that the task is solved.

D. Data-Scaling Law

An earlier scaling sweep (val/test fixed) suggested an early plateau: $\approx 96\%$ of the recorded FG-NET large-gap gain at 10% of identities ($\approx 1,500$), reaching a plateau from 25% (Supplementary Table S-V and figure; each point is mean $\pm$ std over three seeds); the slight dip at full data (0.50 $\rightarrow$ 1.00) lies within seed variance—the curve is flat, not monotonic, beyond $\approx 10\%$. Because intermediate-fraction checkpoints were overwritten in that sweep, these values remain exploratory until the schema-v2 per-fraction rerun; no data-efficiency claim depends on them.

E. Endpoint-Age-Matched Internal Check

On a stricter internal 25+ protocol, each negative retains the positive anchor and replaces only the counterpart with a different person’s face of matching age (within 2 years). All 154 positives have matched negatives; age gap alone gives ROC-AUC 0.500. FaceNet improves from 0.768 to 0.833 ROC-AUC (paired subject-bootstrap $\Delta = +0.065$, 95% CI [+0.005, +0.130]), but TAR@FAR=1% falls from 0.325 to 0.253. Thus the earlier age-bucket result (0.517 frozen AUC) overstated failure of the frozen model; these data support a bounded AUC gain, not elimination of an age shortcut or an across-operating-point improvement. A linear age probe still finds age information in the embedding (Table XIII); whether the gain reflects a changed decision geometry requires a separate mechanism test.

F. Tested Objectives and Controlled Method Comparisons

We train the canonical modern-FR margin objectives—ArcFace-, CosFace- and SphereFace-margin identity classification [17] (the core of OE-CNN/MTLFace)—on our naturally-supervised identities (9,204 classes), same weak backbone,

scope and protocol. On the original random-impostor FG-NET large-gap protocol the contrastive, ArcFace and CosFace objectives have similar point estimates (0.848 / 0.851 / 0.843, within ± 0.006 ; Table XIV; plotted in the supplement); only SphereFace—the hardest margin to optimize, here without λ -annealing on 2–3-shot identities—trails (0.801). We conclude only that the tested objectives perform similarly on these mined pairs; a fully matched-source control is needed to attribute the gain causally to the data source. ArcFace/CosFace additionally *forget less* on easy benchmarks—a practical recommendation. A *noise-robust* comparator—sub-center ArcFace [21], which keeps $K=3$ centroids per identity so that label-noisy or low-quality crops route to off-centers—has the same recorded large-gap 0.851 (our.25+ 0.869); this comparison shows no observed point-estimate advantage, not equivalence or immunity to label noise. On corrected endpoint-age-matched FG-NET 25+, three-seed ArcFace-R50/CASIA reimplementations yield MTLFace 0.8327 ± 0.0012 and CACon 0.7995 ± 0.0027 versus frozen 0.8180. MTLFace’s seed-42 gain interval includes zero; the paired CACon-minus-MTLFace interval also includes zero for one seed. Thus their mean ranking is descriptive, not robust superiority across subjects. CACD-VS AUC is 0.9920 ± 0.0000 versus 0.9907 ± 0.0001 . The supplement separates corrected subject-level inference from historical random-impostor results; these controlled reimplementations are not published leaderboard systems.

Partial source control. With the same VK source, backbone, initialization, 500 persons, 1,096 positive images, 600 positive/600 age-matched negative training pairs and epoch fixed at 10, large-gap (25+ years) pairs exceed 1–2-year pairs on the corrected FG-NET endpoint by $+0.0257 \pm 0.0009$ ROC-AUC over three seeds; each paired subject-bootstrap CI excludes zero (Supplement). This is a within-source association: total training images differ (LOW 1,398, CROSS 1,121), endpoint age/quality and selection remain unmatched, and generic/noise comparative results are absent.

G. Apparent-Demographic Audit

Stratifying by *apparent* gender/age of the pair anchor (estimated, *not* self-identified), no stratified group degrades and every gain is significant (95% paired-bootstrap CI > 0 ; Supplementary Table S-VI and figure). The largest gain is for the youngest group 0–17 (frozen 0.750 $\rightarrow$ +pairs 0.880, +0.130, CI not overlapping adult bands)—the method strengthens where the base model is weakest (the child $\leftrightarrow$ adult regime), narrowing the apparent-demographic gap. A related check uses *non-VK* data but shares the primary FG-NET corpus: on a child $\leftrightarrow$ adult subset of FG-NET (one photo under age 13, the other over 25; 195 positives, 195 negatives), +pairs raises ROC-AUC from 0.684 [0.63, 0.74] to 0.813 [0.77, 0.86] (**+0.129, non-overlapping 95% CIs**), closely matching the internal +0.130. Beyond AUC, an error decomposition at a fixed operating point (per-model global threshold at 1% FMR) shows a genuine reduction in matching errors where it matters: +pairs lowers the false-non-match rate in *every* apparent stratum at matched $\approx 1\%$ FMR—0–17 **0.852 $\rightarrow$ 0.656**, apparent-female 0.674 $\rightarrow$ 0.509, apparent-male 0.743 $\rightarrow$ 0.534—with no stratum

TABLE X
HEADROOM: FG-NET LARGE-GAP BY BACKBONE STRENGTH (CURATED DATA, LR 3E-5).

Backbone (strength)	Frozen	+pairs	Δ large-gap	Δ our.25+
FaceNet (weak)	0.736	0.848	+0.112	+0.198
ArcFace r50 (weak, diff. arch.)	0.764	0.805	+0.041	+0.133
AdaFace IR-50 (strong)	0.917	0.924	+0.007	+0.004
ArcFace r100 (strong)	0.954	0.915	−0.039	−0.002
AdaFace IR-101 (strong)	0.957	0.931	−0.026	−0.006

TABLE XI
AGE-LEAKAGE LINEAR PROBE (CURATED DATA; CHANCE = 0.25).

Model	Age-probe bal. acc. ↓	Identity ROC-AUC ↑
frozen	0.651 ± 0.019	0.836
+pairs	0.629 ± 0.012	0.916
+pairs+disentangle	0.632 ± 0.017	0.918

TABLE XII
OBJECTIVE COMPARISON (FACE NET, CURATED DATA, SEED 42): CONTRASTIVE, ARC FACE AND COS FACE HAVE SIMILAR POINT ESTIMATES UNDER THE ORIGINAL RANDOM-IMPOSTOR FG-NET LARGE-GAP PROTOCOL, NOT DEMONSTRATED EQUIVALENCE; SPHERE FACE (NO λ -ANNEALING) TRAILS ON OUR SPARSE FEW-SHOT IDENTITIES. THE 0.838 INTERNAL AUC IS SINGLE-SEED; THE SUPPLEMENT REPORTS 0.8348 AS A HISTORICAL THREE-SEED MEAN.

Objective	FG-NET l.g.	our.25+	LFW	AgeDB-30
Frozen	0.736	0.640	0.969	0.953
+pairs (contrastive)	0.848	0.838	0.946	0.945
+ArcFace	0.851	0.868	0.952	0.947
+ArcFace, sub-center	0.851	0.869	0.951	0.947
+CosFace	0.843	0.860	0.952	0.945
+SphereFace	0.801	0.699	0.950	0.943

regressing, and per-stratum score calibration (logistic slope 9.0–11.7) stays uniformly positive and stable; the only exception is the small 45+ stratum ($n \approx 150$), whose FMR scatters to 2.1% at the global threshold (sampling noise). We deliberately avoid the word “fair” as a solved property: the source is apparent-female-skewed (76.7%), attributes are apparent (not ethnicity or legal categories), and 45+ is sparse.

H. Cross-Source Transfer

Training on one community and testing on the held-out other transfers the gain: external FG-NET large-gap +0.113 (vs. +0.112 within-source—essentially identical) and internal +0.092 on the held-out community. We frame this as transfer between two independent sources within one platform domain; Sec. V-I extends the test to an independent non-VK platform, and we do not claim web-scale generalization.

I. Cross-Platform Transfer (Independent, Non-VK Source)

A key external check of a naturally-supervised *source* is whether it generalizes beyond the originating platform. We apply the *identical* mining protocol to an independent, non-VK community—the Reddit board r/PastAndPresentPics (same

TABLE XIII
CROSS-PLATFORM TRANSFER: A VK-TRAINED MODEL ON AN INDEPENDENT REDDIT (NON-VK) THEN/NOW TEST SET (FROZEN VS. +PAIRS, TRAINED ON VK ONLY; 396 POSTS → 1,575 POSITIVE PAIRS AFTER THE MULTI-PERSON-COLLAGE FILTER).

Reddit (non-VK) test metric	Frozen	+pairs (VK-trained)
Overall ROC-AUC	0.718	0.757±0.008
EER	0.345	0.323±0.005
TAR@FAR=1%	0.271	0.285±0.012
Test pairs (pos / neg)	1,575 / 1,575	

TABLE XIV
CROSS-SOURCE TRANSFER IN BOTH DIRECTIONS (MEAN±SD OVER THREE FINE-TUNING SEEDS). REDDIT→VK USES A SMALL 220-PERSON TRAINING SOURCE AND IS SUPPORTING RATHER THAN HEADLINE EVIDENCE.

Train	Test	Frozen	Tuned	Δ
VK	Reddit (overall)	0.718	0.757±0.008	+0.039
Reddit	FG-NET large-gap	0.736	0.779±0.004	+0.043
Reddit	VK internal 25+	0.640	0.657±0.002	+0.016

“then/now” format, same LLM parser)—and evaluate three VK-trained +pairs models (*no* Reddit data in training) on 1,575 Reddit pairs (Table XVI): overall ROC-AUC rises **0.718** → **0.757±0.008**; the maximum seed-wise 95% CI width is 0.033. EER falls from 0.345 to 0.323±0.005. The transfer is *bidirectional* (Table XVII): three weak-backbone runs trained on *only* the small 220-person Reddit source also help on FG-NET and VK. Reddit then/now pairs are inherently cross-age, so the frozen 0.718 sits far below its easy-benchmark ceiling, mirroring the within-VK large-gap regime; the gain is smaller than within-VK (a noisier, partly collage-derived set), so we read it as evidence of cross-platform/source transfer, not web-scale or cross-cultural generalization and not a SOTA claim. The explicit ≥ 25 -year Reddit subset has only ten positives and is therefore not interpreted separately. Alongside our other external evaluations (plotted in the supplement), the +pairs gain stays *positive across all three*, shrinking on harder/noisier sets.

J. Calibration

Raw cosine (Platt) is mis-calibrated across gap bins—over-confident at 15–25 years (ECE 0.052); a $P(\text{same} \mid \text{cosine, age-gap})$ calibrator fixes every bin (15–25 → 0.017) and improves the overall Brier score (0.035 → 0.029)—a usable, age-aware same-identity probability.

K. Disentanglement

Explicit age removal (gradient reversal + age head, MTLFace-style) adds only a small, stable cross-age gain over +pairs (+0.006 FG-NET large-gap, +0.015 our.25+) without harming easy benchmarks; its absolute large-gap (≈ 0.853) matches the ArcFace objective (0.851), showing only that this tested add-on gives little further improvement under the same data and protocol.

L. Hard-Negative Mining

Adding the hardest cross-person negatives to fine-tuning—the top-5 most-similar other-person faces per anchor (cosine in a hard band, excluding the near-duplicate/uncertain range)—further sharpens large-gap discrimination: FG-NET large-gap rises to 0.868 ± 0.004 over three seeds (from 0.848 with random negatives), but easy benchmarks forget *more* and FG-NET overall ROC actually drops (LFW 0.969 $\rightarrow$ 0.925, AgeDB-30 0.953 $\rightarrow$ 0.911, CALFW 0.948 $\rightarrow$ 0.898, FG-NET ROC 0.923 $\rightarrow$ 0.912; Supplementary Table S-VII). Hard negatives thus trade general-benchmark accuracy for large-gap separation—useful for a cross-age retrieval setting, not for a general verifier. (A secondary result, not part of the headline; the internal hard-negative test is near-chance for the frozen model by construction and is not comparable to the standard internal test.)

VI. DISCUSSION

The contribution is a naturally-supervised source whose mined pairs improve the corrected large-gap FG-NET endpoint for the tested weak recognizer. Holding architecture fixed does not separate longitudinal supervision from all other properties of the training data. The historical LOW/CROSS control has unequal total image budgets; its restricted-budget replacement and generic/noise controls remain incomplete. Similar point estimates across tested objectives are not an equivalence test, and the tested FRAN controls do not represent synthesis methods in general. The effect is not a uniform upgrade: AgeDB-30 declines in the historical pair-level analysis, CACD-VS loses 10-fold accuracy, and the corrected internal test loses low-FAR TAR despite gaining AUC. The separate source-bound LFW evaluation supports a fixed-checkpoint AUC gain, but its accuracy, EER and low-FAR gain intervals include zero in the supplement; Table VIII instead reports historical comparisons. Thus historical aligned-cache operating points must not substitute for source-bound uncertainty or deployment-calibrated error rates.

A. Scope of the Claims

To forestall over-reading, we state explicitly what the evidence does and does not establish.

Observed (conditional on current weights and protocols). (i) FaceNet improves endpoint-age-matched FG-NET large-gap ROC-AUC over three checkpoints, with positive per-checkpoint paired subject intervals on the same subjects; (ii) the tested source-community transfers and child $\leftrightarrow$ adult FG-NET subset provide supporting evidence, not independent replications of

the main FG-NET corpus; (iii) real pairs outperform the tested FRAN controls under the historical protocol; (iv) contrastive, ArcFace and CosFace objectives have similar historical point estimates, without establishing equivalence; (v) the internal endpoint-matched weak-backbone probe gains AUC but low-FAR TAR declines, and age remains decodable; (vi) curation removes roughly half of the raw positive-pair count as near-duplicate frames.

Not established / out of scope. (a) causal superiority of longitudinal data over fully matched generic or short-gap controls; (b) a uniform operating-point gain, strong-backbone mechanism or objective equivalence; (c) web-scale, population-wide or cross-ethnicity transfer from the small VK-Reddit study; (d) human-validated supervision precision, completed train-benchmark overlap adjudication or complete legacy training provenance. Subject intervals are conditional on evaluated checkpoints, not a population of training seeds; unresolved person metadata leaves other benchmark intervals pair-level.

VII. LIMITATIONS

- **Backbone dependence:** an exploratory one-seed curve concentrates the gain in weak/medium backbones and shows null/negative changes for strong models. The planned fixed-budget multi-seed mechanism matrix is incomplete, so this is a provisional scope bound, not a saturation claim. Look-alike negatives challenge a task-level ceiling interpretation (Table X); their identity purity also awaits the blinded audit.
- **External validity:** two VK communities, one platform, one language/cultural context; apparent-female-skewed (76.7%); sparse at apparent age 45+. Transfer is shown between two VK sources, to standard external benchmarks, and to a small independent Reddit then/now test; this does not establish platform-agnostic generalization.
- **Apparent attributes only** in the fairness audit (estimated, not self-identified gender/ethnicity).
- **Naturally-supervised, not strictly label-free:** an LLM parses ages and validates group integrity; clustering forms persons—audited (Sec. III-C) but a residual hidden-noise source.
- **Statistics:** seed variance, paired bootstrap (fairness), and benchmark CIs / EER / TAR@FAR (Table VIII; the internal identity-level interval is reported in text) are reported in-text; DeLong tests for correlated ROC curves, an FMR/FNMR error decomposition and per-stratum calibration are now reported (Sec. V-A, Sec. V-G); a full per-cell multiple-comparison correction across every benchmark $\times$ operating-point remains for the release supplement.
- **Benchmark scale and age:** CACD-VS (4,000 pairs), AgeDB-30 and CALFW (6,000 each) complement FG-NET, but do not replicate its ≥ 25 -year endpoint. The primary large-gap effect therefore remains dependent on small FG-NET and partial detector coverage; its child $\leftrightarrow$ adult analysis shares that corpus. The internal test and small Reddit transfer are supporting evidence, not independent confirmation of the same endpoint. Larger independent longitudinal data and verified benchmark person metadata remain necessary for stronger claims.

- **Controlled SOTA scope:** three-seed common-protocol MTLFace and CACon reimplementations fix the ArcFace-R50 backbone, person split and compute budget while retaining their method-specific objectives. They test method behavior under our resource/data regime, not exact reproduction of the much larger published systems. Non-comparable leaderboard results remain in a separate supplement table.

VIII. ETHICS, LEGAL BASIS AND DATA GOVERNANCE

This is a sensitive biometric study; we treat governance as a first-class component rather than as a disclaimer. Permitted use is restricted to controlled / consent-based / human-reviewed applications (rights-cleared archives and family photos, authorized humanitarian search, account recovery); the system outputs candidates for human review, never an automatic decision. This is a *research-only* study—nothing is deployed, no automated decision is made, and *no public biometric database is released* (Sec. VIII-B). Forbidden: mass identification, de-anonymization, surveillance, and any processing of minors' data without a legal basis.

A. Ethical Approval and Legal Basis

Ethical approval. The study was not placed under institutional review board oversight before the retrospective analysis. There is no interaction or intervention with any person: the study analyses images previously posted on public community walls and retrieved via the platform's official API. Nobody is contacted or ranked, and nothing is deployed. However, the images are biometric, the people remain identifiable, individual consent was not obtained, and images depicting minors were included in aggregate research analyses. We therefore do *not* self-declare the work exempt. A formal retrospective determination from our university's ethics committee—explicitly covering biometric processing, minors, manual annotation, controlled access and publication—is required before resubmission; the final article and submission form will report the committee's actual decision, name, date and reference. No released artifact contains raw faces, raw posts or recoverable platform identifiers (Sec. VIII-B).

Data were collected from public “then/now” community walls via the platform's official API. We stress that *official API access and voluntary public posting do not by themselves authorize biometric processing*. The images remain identifiable, explicit consent was not obtained, and the applicable legal basis and safeguards have not yet been confirmed. Scientific-research provisions may be relevant, but this draft does not self-assign a lawful basis or claim full compliance; the committee determination and specialist data-protection review must decide whether the retrospective corpus, minors and any controlled access can be retained.

B. De-identification and Release Policy

We *do not release raw face images or raw posts*. The public repository is restricted to code, configurations and aggregate statistics. A draft controlled-access procedure and DUA exist, but no trained weights, embeddings, crops, captions

or pseudonymized benchmark rows are currently offered: embeddings remain biometric templates, and salted hashes are pseudonyms rather than anonymization. Any future access route, permitted fields and deletion workflow are conditional on the institutional determination and data-protection review. Paper figures contain no identifiable faces.

C. DPIA and Minimization

Because the processing is biometric and large-scale, the final governance package requires a Data Protection Impact Assessment addressing re-identification, function creep and minors. Its completion and approval are publication gates; this draft does not claim that review has occurred. Data minimization excludes names and social graphs from derived research tables.

D. Minors

The child↔adult regime necessarily involves images depicting minors and therefore requires heightened review rather than an author-supplied legal conclusion. Such items were included in aggregate, non-redistributed research analyses but are withheld from every release. Their continued use, any further annotation, and any claim derived from them are conditional on the formal institutional determination described above; otherwise they will be removed and all affected results recomputed.

E. Subject Rights

A takedown/deletion workflow is drafted so that a per-item status can propagate to derived artifacts. It, the datasheet [15], DUA and DPIA must be approved before any controlled access begins.

IX. REPRODUCIBILITY

We provide the non-sensitive scripts and configurations; full hyperparameters are versioned. Key choices: *usable-face* filtering (det-score, size, Laplacian blur, single target); detector/alignment InsightFace buffalo_l RetinaFace + 5-point ArcFace norm_crop (112 px; FaceNet input 160 px); ArcFace w600k_r50 embeddings for *clustering* (union-find, cosine ≥ 0.85) and *dedup* (cosine ≥ 0.97); GigaChat with versioned prompts and cached outputs; a *by-person* 70/15/15 split (seed 42, balanced negatives, plus a look-alike variant); margin-contrastive training with seeds 42/1/2; and machine-readable manifests containing commands, checksums, package versions and metrics. No claim is made that the private biometric inputs are independently available before the access route is approved.

Compute. The entire study runs on a single laptop: one NVIDIA RTX 3060 Laptop GPU (6 GB), an AMD Ryzen 7 5800H (8 cores / 16 threads) and 16 GB RAM under Windows 11; the software stack is Python 3.12, PyTorch 2.12 (CUDA 13.2, cuDNN 9.2), with InsightFace 1.0 / ONNX Runtime 1.26 for detection and embeddings and scikit-learn 1.9 / Matplotlib 3.11 for analysis and figures. The weak-backbone protocol keeps the full study within a 6 GB GPU budget.

Data availability. Code, configurations and aggregate manifests are public; raw faces, posts, captions, embeddings, weights derived from the private corpus, and row-level protocols are not. Consequently, the data-mining stage is not presently independently reproducible. The final statement will name only an access mechanism actually approved by the committee, data-protection review and venue; otherwise results dependent on unavailable data will be removed from the submission.

X. CONCLUSION AND FUTURE WORK

Mined multi-photo posts provide a weakly/naturally-supervised training signal. On the corrected endpoint-age-matched FG-NET protocol, fine-tuning the tested weak recognizer improves large-gap ROC-AUC with positive per-checkpoint subject intervals; the checkpoints share the same evaluation subjects. This is evidence for fine-tuning on these pairs, not yet for the added value of longitudinal pairing over fully matched data. Historical objective comparisons show similar point estimates, not equivalence, and FRAN controls support only a method-specific synthesis comparison. External results are mixed: AgeDB-30 declines, CACD-VS accuracy declines, and source-bound LFW supports AUC improvement but not a reliable accuracy or low-FAR gain. The corrected internal test likewise gains AUC while losing low-FAR TAR. Common-protocol MTLFace/CACON reimplementations and small three-seed VK-Reddit transfers extend the evaluation without proving leaderboard superiority or population-wide transfer. Before resubmission, matched-data controls, the strong-backbone matrix, legacy training provenance, blinded identity/annotation audits and the formal institutional determination must be completed; claims must follow their actual outcomes.

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REFERENCES

- [1] Y. Wang, D. Gong, Z. Zhou, X. Ji, H. Wang, Z. Li, W. Liu, and T. Mei, "Orthogonal deep features decomposition for age-invariant face recognition," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 738–753.
- [2] H. Wang, D. Gong, Z. Li, and W. Liu, "Decorrelated adversarial learning for age-invariant face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 3527–3536.
- [3] Z. Huang, J. Zhang, and H. Shan, "When age-invariant face recognition meets face age synthesis: A multi-task learning framework," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 7282–7291.
- [4] B.-C. Chen, C.-S. Chen, and W. H. Hsu, "Cross-age reference coding for age-invariant face recognition and retrieval," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 768–783.
- [5] K. Ricanek and T. Tesafaye, "MORPH: A longitudinal image database of normal adult age-progression," in *Proc. IEEE Int. Conf. Autom. Face Gesture Recognit. (FG)*, 2006, pp. 341–345.
- [6] S. Moschoglou, A. Papaioannou, C. Sagonas, J. Deng, I. Kotsia, and S. Zafeiriou, "AgeDB: The first manually collected, in-the-wild age database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2017, pp. 51–59.
- [7] G. Panis, A. Lanitis, N. Tsapatsoulis, and T. F. Cootes, "Overview of research on facial ageing using the FG-NET ageing database," *IET Biometrics*, vol. 5, no. 2, pp. 37–46, 2016.
- [8] H. Wang, V. Sanchez, and C.-T. Li, "Cross-age contrastive learning for age-invariant face recognition," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, 2024, arXiv:2312.11195.
- [9] —, "From age estimation to age-invariant face recognition: Generalized age feature extraction using order-enhanced contrastive learning," *arXiv preprint arXiv:2501.01760*, 2025.
- [10] W. Yao, M. A. Farooq, J. Lemley, and P. Corcoran, "Synthetic face ageing: Evaluation, analysis and facilitation of age-robust facial recognition algorithms," *arXiv preprint arXiv:2406.06932*, 2024.
- [11] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 1597–1607.
- [12] T. Zheng, W. Deng, and J. Hu, "Cross-age LFW: A database for studying cross-age face recognition in unconstrained environments," *arXiv:1708.08197*, 2017.
- [13] G. B. Huang, M. Ramesh, T. Berg, and E. Learned-Miller, "Labeled faces in the wild: A database for studying face recognition in unconstrained environments," Univ. of Massachusetts, Amherst, Tech. Rep. 07-49, 2007.
- [14] J. Deng, J. Guo, E. Ververas, I. Kotsia, and S. Zafeiriou, "RetinaFace: Single-shot multi-level face localisation in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 5203–5212.
- [15] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford, "Datasheets for datasets," *Commun. ACM*, vol. 64, no. 12, pp. 86–92, 2021.
- [16] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 815–823.
- [17] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 4690–4699.
- [18] M. Kim, A. K. Jain, and X. Liu, "AdaFace: Quality adaptive margin for face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 18 750–18 759.
- [19] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.
- [20] G. Zoss, P. Chandran, E. Sifakis, M. Gross, P. Gotardo, and D. Bradley, "Production-ready face re-aging for visual effects," *ACM Trans. Graph. (SIGGRAPH Asia)*, vol. 41, no. 6, pp. 1–12, 2022.
- [21] J. Deng, J. Guo, T. Liu, M. Gong, and S. Zafeiriou, "Sub-center ArcFace: Boosting face recognition by large-scale noisy web faces," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 741–757.
- [22] European Parliament and Council of the European Union, "Regulation (eu) 2016/679 (general data protection regulation)," *Official Journal of the European Union*, L119, 2016.